MCS 320 Quiz 2 Friday 4 September 2026

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. Use the first 10 convergents of the continued fraction of $\cos(\sqrt{11})$ to define the first 10 rational approximations of $\cos(\sqrt{11})$.

   Compute the relative errors of those 10 rational approximations to verify the accuray.

2. Let $p = x^2 - x - 1$ be a polynomial with rational coefficients.

   Add sufficiently many roots to the field of rational coefficients so that p can be written as the product of two linear factors.